CYBERSECURITY HACKATHON

Team Aegis- Track 1

AGENTIC AI-POWERED IOT SECURITY SYSTEM

A microcontroller (laptop for the prototype) acts as the autonomous traffic supervisor, orchestrating four specialized AI agents with deterministic task scheduling and <100ms response latencies.

MCU specifications-

Hardware: ESP32-S3 dual-core 240 MHz Xtensa or Raspberry Pi 4B (1.5 GHz Cortex-A72)

Task scheduling (100ms tick-based execution), Inter-agent message routing, Sensor fusion (video + network + firmware data), Automated threat response triggering, Dashboard data serialization (So transfer of information between agents is easy), Watchdog monitoring for agent health.

INTRODUCTION

FOUR AUTONOMOUS AI AGENTS

DEVICE DISCOVERY

Agent 1 performs continuous device discovery and profiling by analyzing ARP traffic, Nmap scans, DNS, and mDNS data. It fingerprints devices, identifies firmware and open ports, and builds an inventory. Using heuristics and CVE databases, it flags risks, completing each discovery cycle within 500 milliseconds every 30 minutes automatically securely.

ANAMOLY DETECTION

Agent 2 establishes normal network behavior for each device by analyzing traffic flows such as IPs, ports, protocols, and durations. It learns baselines using unsupervised training and detects deviations in real time. Using a quantized Isolation Forest model, it assigns anomaly scores and flags suspicious flows within 50 milliseconds.

VIDEO ANALYTICS

Agent 3 analyzes live IP camera video to detect objects, estimate human poses, and classify behavior in real time. It generates intrusion alerts, activity timelines, and behavior heatmaps. Using lightweight YOLO, OpenPose, and LSTM models optimized with TFLite, it processes each frame end-to-end within 250 milliseconds.

DECEPTION INTELLIGENCE

Agent 4 uses deception to identify and profile attackers by monitoring failed logins, port scans, and malicious payloads. It deploys lightweight SSH, HTTP, and MQTT honeypots to attract threats. The agent generates real-time alerts and attacker behavioral signatures, responding to connection attempts in under 100 milliseconds.

DEVICE DISCOVERY

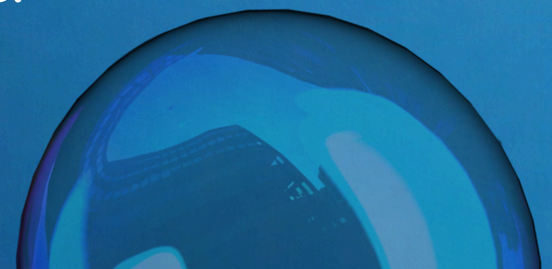
Device discovery uses **ARP packets** and **Nmap scans** to identify all devices connected to a network.
ARP requests reveal active hosts by mapping IP addresses to MAC addresses when devices respond. Nmap then scans these discovered IPs to confirm live hosts, identify open ports, and infer device types, creating an accurate inventory of connected devices on the network.

ANAMOLY DETECTION

The anomaly score is calculated in two stages: a base risk estimation and a contextual final score.

First, a base score is derived from observed behavior such as scans on vulnerable ports (e.g., Telnet 23, SMB 445, RDP 3389, FTP 21) and repeated access patterns. High-risk ports and frequent identical patterns increase this score, capped to avoid over-penalization.

Second, the final anomaly score combines this base value with the assessed risk level (low/medium/high), the number of suspicious signals detected (e.g., botnet indicators, unusual login times), and a small random factor to prevent uniform scoring. The result is normalized between 0 and 1, producing a realistic, interpretable anomaly score.



ANOMALY SCORE CALCULATION

```
def final_score(base, risk, signals):
    score = base
#from gemini prompt
    score += {"low": 0.1, "medium": 0.2, "high": 0.4}.get(risk, 0.2)

#no. of alerts issues for a particular ip (like nighttime logging in, botnet alerts etc.)
    score += 0.05 * len(signals)

# break uniformity
    score += random.uniform(-0.05, 0.05)

    return round(min(score, 1.0), 2)
```

ANOMALY SCORE CALCULATION

```
#risk score due vulnerable ports

def base_score(ports, same_pattern):
    score = 0.1
    if 23 in ports or 445 in ports:
        score += 0.4
    if 3389 in ports:
        score += 0.4
    if 21 in ports:
        score += 0.2
    if same_pattern > 5:
        score += 0.2
    return min(score, 0.6)
```

VULNERABLE PORTS

21, FTP: Plain-text credentials, brute-force, anonymous login

23, Telnet: No encryption, easy credential sniffing

445, SMB: Ransomware (WannaCry), file sharing exploits

3389, RDP: Brute-force, credential stuffing

and so on..

DECEPTION INTELLIGENCE

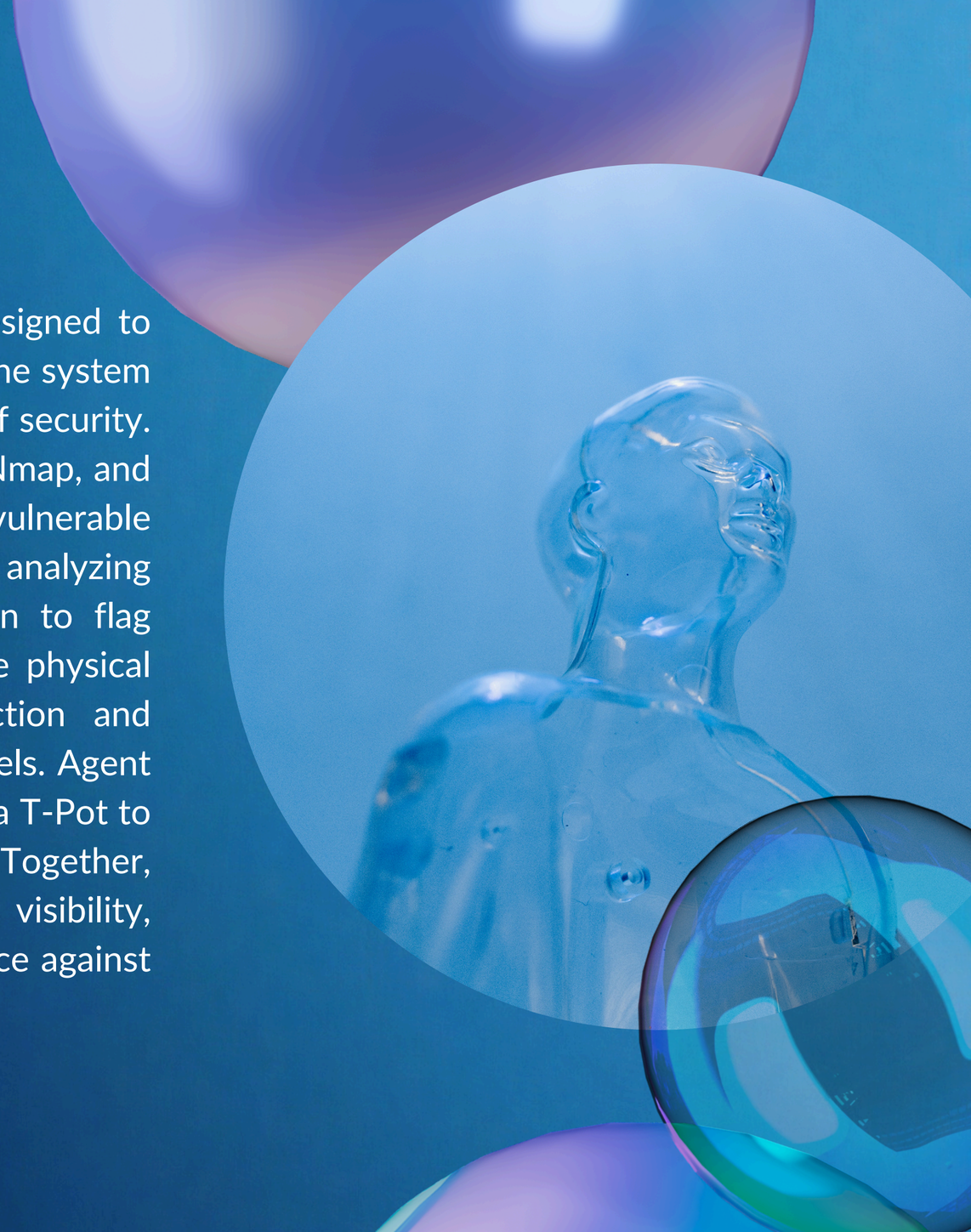
For Agent 4, we selected T-Pot as the core platform for deception and threat intelligence due to its ability to deploy and manage multiple honeypots in a unified, secure environment. T-Pot integrates diverse low- and medium-interaction honeypots, enabling broad coverage across common attack vectors such as SSH, HTTP, and MQTT. This makes it well suited for detecting reconnaissance activity, credential brute-force attempts, and automated scans targeting IoT and networked systems.

In our architecture, honeypots are dynamically engaged after repeated failed login attempts, abnormal port scanning behavior, or suspicious payload patterns are detected. This approach ensures deception is applied selectively, reducing noise while maximizing the capture of meaningful attacker interactions.

T-Pot's centralized logging and visualization capabilities allow us to correlate attacker IPs, behaviors, and timelines, transforming raw interaction data into actionable threat intelligence. By isolating these honeypots from production assets, Agent 4 safely studies adversary techniques, supports attacker profiling, and strengthens overall situational awareness without increasing operational risk.

SUMMARY

This project presents an agent-based intelligent security framework designed to monitor, detect, and respond to cyber and physical threats in real time. The system is composed of four coordinated agents, each addressing a critical layer of security. Agent 1 performs continuous device discovery and profiling using ARP, Nmap, and network metadata to maintain an up-to-date inventory and identify vulnerable assets. Agent 2 establishes normal behavioral baselines for each device by analyzing network flows and applies machine learning–based anomaly detection to flag deviations indicative of compromise. Agent 3 extends security into the physical domain through real-time video analytics, enabling intrusion detection and behavioral profiling using lightweight, edge-optimized deep learning models. Agent 4 focuses on deception and threat intelligence by deploying honeypots via T-Pot to attract attackers, capture malicious activity, and build attacker profiles. Together, these agents create a proactive, layered defense system that combines visibility, intelligence, and deception to enhance situational awareness and resilience against evolving threats.



THANK YOU